

AP COMPUTER SCIENCE PRINCIPLES

Our AP CSP Learning Story

These district Learning Intentions guide the course. They describe three capacities that overlap, recur, and develop together as students become computational thinkers.

**LI 1 INVESTIGATE SYSTEMS**

I am learning to investigate computing innovations and apply computational thinking to explain how a system works.

LI 2 CREATE PROGRAMS

I am learning to create a computer program.

LI 3 ANALYZE & SOLVE

I am learning to analyze computational artifacts and develop solutions to computing problems.

COMPUTATIONAL THINKING: THE PRACTICES College Board's six practices describe what computational thinkers do; the Learning Story shows how those practices grow through the three district intentions.

COMPUTATIONAL THINKING PRACTICE	WHAT IT LOOKS LIKE	STRONG CONNECTIONS
P1 COMPUTATIONAL SOLUTION DESIGN	Investigate a task, define its purpose, and design and evaluate possible approaches.	LI 1 LI 2 LI 3
P2 ALGORITHMS & PROGRAM DEVELOPMENT	Represent, develop, implement, and apply algorithms to create a solution.	LI 2 LI 3
P3 ABSTRACTION IN PROGRAM DEVELOPMENT	Use variables, procedures, and data abstractions to manage complexity.	LI 1 LI 2 LI 3
P4 CODE ANALYSIS	Explain, trace, test, evaluate, and correct algorithms and programs.	LI 1 LI 2 LI 3
P5 COMPUTING INNOVATIONS	Investigate how systems, data, and innovations work and affect people.	LI 1 LI 3
P6 RESPONSIBLE COMPUTING	Collaborate and contribute to safe, inclusive, secure, and ethical computing.	LI 1 LI 2 LI 3

HOW TO READ THE TILES Each rounded tile on the next page is one Success Criterion. Its numbered badges show every Learning Intention it supports. One piece of evidence may demonstrate several criteria.

OUR AP CSP LEARNING STORY

Success Criteria

Each tile is one observable ability. The numbered badges show every district Learning Intention it supports.

1 Investigate systems

I can evaluate data for accuracy, relevance, sufficiency, representativeness, and bias, and explain how limitations affect claims or outcomes.

1 3

I can encode, decode, and explain how bits and binary patterns represent numbers, text, color, images, and analog information.

1 3

I can explain how addressing, packets, routers, redundancy, fault tolerance, and protocols work together to move data across the Internet.

1 3

I can create an interactive program that integrates HTML, JavaScript, DOM elements, media, and styling to serve a purpose.

1 2 3

I can decompose a program into reusable procedures with parameters and return values and explain how abstraction manages complexity.

1 2 3

I can implement and compare search and sort algorithms, explaining their requirements, behavior, and efficiency.

1 2 3

2 Create programs

I can clean, organize, visualize, and interpret data to identify patterns and distinguish correlation from causation.

1 3

I can evaluate tradeoffs in digital storage and representation, including precision, overflow, round-off, and lossless or lossy compression.

1 3

I can investigate a computing problem or task, define its purpose and constraints, and compare possible solution approaches.

1 2 3

I can use variables, data types, operators, input, and output to represent, transform, and communicate program data.

2 3

I can use events and timers to create responsive program behavior.

2 3

I can trace code, predict results, design test cases, identify and correct errors, and refine a program using evidence and feedback.

1 2 3

3 Analyze & solve

I can explain how big data, crowdsourcing, parallel computing, and machine learning generate knowledge and evaluate their benefits, limitations, and bias.

1 3

I can explain how devices and physical media form networks and how bit rate, bandwidth, and latency affect performance.

1 3

I can collaborate, use feedback, document decisions, and credit others while developing a computational solution.

2 3

I can represent and implement algorithms using sequence, selection, iteration, Boolean logic, pseudocode, and flowcharts.

1 2 3

I can create, modify, and traverse arrays or lists to organize data and compute results.

2 3

I can evaluate computing innovations for impacts on people and society and make safe, inclusive, secure, and ethical computing choices.

1 2 3